Interest Rate Spillovers to the US Treasury Curve

5-Country VAR: US, EU, CA, BR, UK

Nelson-Siegel Level, Slope, Curvature

Rates Research | 5 May 2026

Methodology: 5-Country VAR on Nelson-Siegel Factors

We decompose five government yield curves (US, EU, CA, BR, UK) into Level, Slope, and Curvature via Nelson-Siegel ($\lambda = 0.10$). A 15-variable VAR(1) is estimated on weekly data (2005–2026, $N = 1112$ obs).

Cholesky ordering: EU $\rightarrow$ CA $\rightarrow$ BR $\rightarrow$ UK $\rightarrow$ US

Rationale: US is placed last as the “receiver” of interest. US is most exogenous (global reserve currency).

Granger Causality Tests ($\alpha = 5\%$)

Source	US Factor	F-stat	p-value	Sig.?
EU	Level	7.07	0.000	**Sig.**
EU	Slope	7.35	0.000	**Sig.**
EU	Curvature	3.41	0.017	**Sig.**
CA	Level	0.85	0.466	Not sig.
CA	Slope	0.33	0.801	Not sig.
CA	Curvature	0.84	0.470	Not sig.
BR	Level	0.54	0.652	Not sig.
BR	Slope	0.97	0.406	Not sig.
BR	Curvature	0.37	0.773	Not sig.
UK	Level	6.70	0.000	**Sig.**
UK	Slope	7.08	0.000	**Sig.**
UK	Curvature	4.15	0.006	**Sig.**

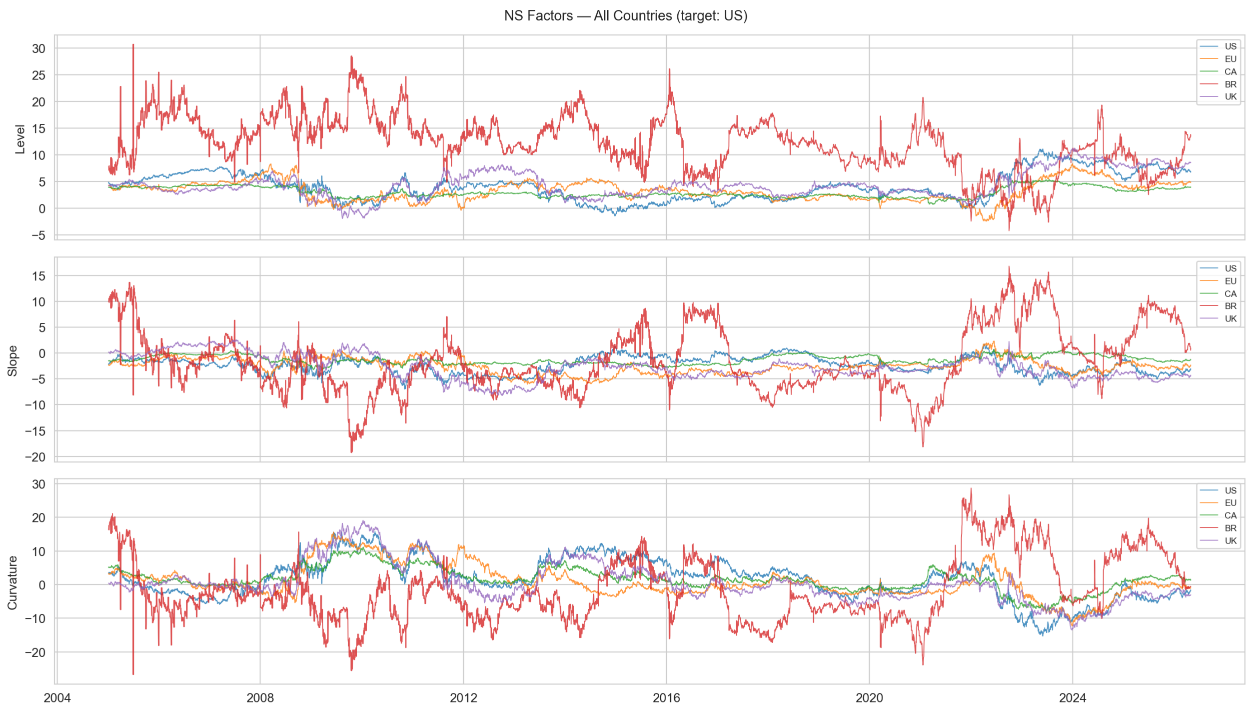

Figure 1: Nelson-Siegel factors for all 5 countries (2005–2026). US highlighted as the target curve.

Shock Propagation to United States

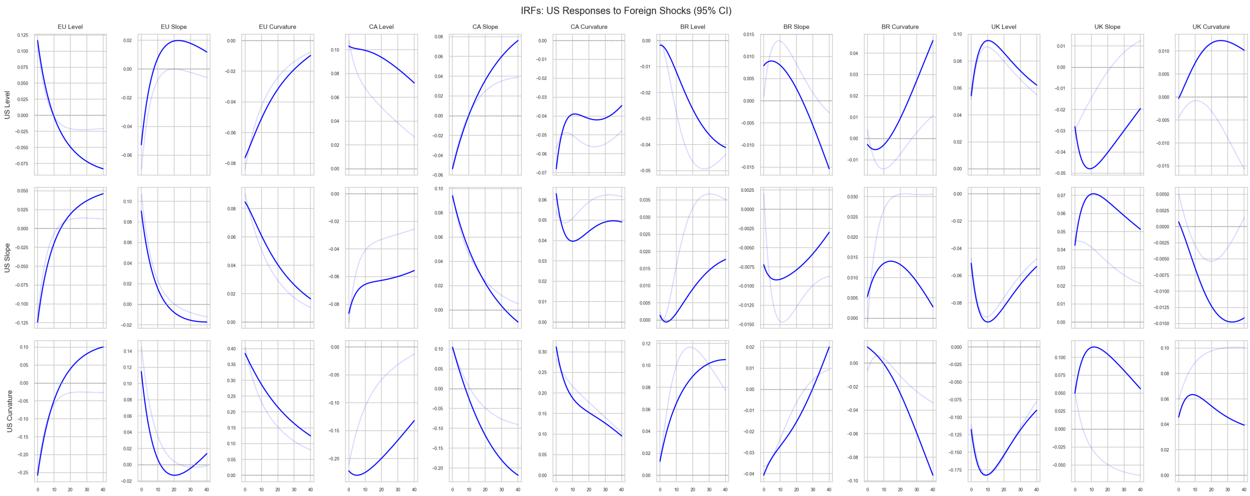

Figure 2: Orthogonalised IRFs: US factor responses to foreign shocks over 40 weeks. Blue shading = 95% bootstrap CI.

FEVD at 52-Week Horizon (% of variance)

	US	EU	CA	BR	UK
US Level	36.1	14.3	28.3	5.0	16.4
US Slope	35.6	14.5	22.4	0.9	26.6
US Curvature	26.9	23.6	33.2	5.2	11.0

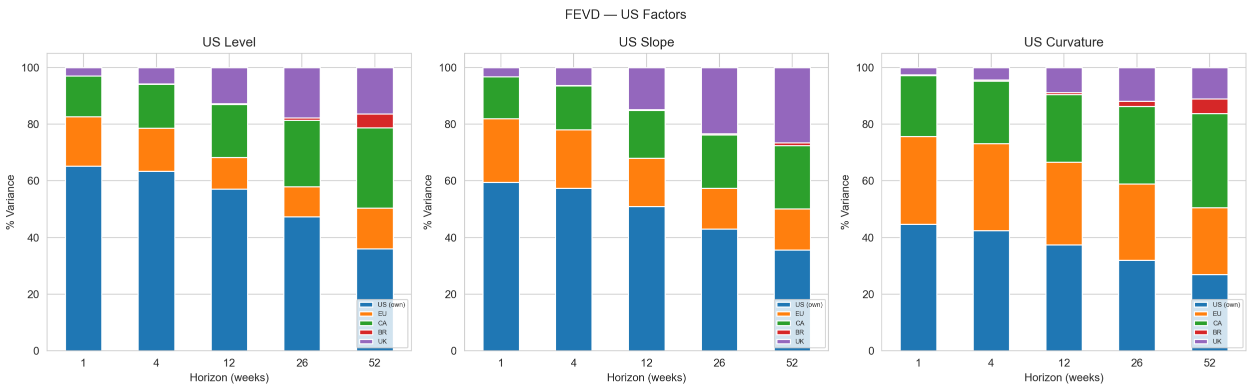

Figure 3: FEVD: share of US factor forecast error variance by source country at horizons 1–52 weeks.

United States: Market Context (May 2026)

Macro & Fiscal Backdrop

- **Inflation:** Core PCE at 2.6% (Mar 2026). Energy pass-through from Iran conflict adding ∼0.5pp to headline.
- **Growth:** Q1 2026 GDP at 1.8% annualised, showing resilience despite tighter financial conditions.
- **Fed:** Held at 4.25–4.50%. Markets no longer pricing cuts in 2026; ∼2 hikes now priced.
- **Issuance:** US Treasury running ∼$2tn annual deficit financing. Refunding mix shifting toward bills.
- **Debt ceiling:** Ongoing uncertainty adds term premium pressure.

Key Drivers Ahead

1. **US is the global anchor:** As the most exogenous rate in our VAR, US shocks propagate to all other curves. Own-variance share (∼36%) is relatively low, reflecting deep global integration.
2. **Canada co-movement:** CA explains 28% of US Level variance—reflecting NAFTA integration and BoC/Fed policy synchronisation.
3. **Fiscal trajectory:** CBO projections of rising debt/GDP create persistent term premium.
4. **Iran ceasefire durability:** Key swing factor for energy pass-through and Fed reaction function.

Data: US: FRED (constant-maturity Treasury rates), EU: ECB Statistical Data Warehouse (AAA-rated), CA: Bank of Canada Valet API (benchmark bonds), BR: IPEA/ANBIMA (DI pre-fixed yield curve), UK: Bank of England IADB (nominal spot curve). Model: VAR(1) on weekly NS factors, sample 2005–2026 (1112 obs).